

Exact Prescribed-Time Prescribed-Accuracy Control for Spacecraft under Dual Reaction Wheel Saturation

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I. Introduction

Increasingly complex spacecraft missions impose stringent requirements on both convergence time and accuracy of attitude control. Especially in time-critical missions such as rendezvous and docking, the spacecraft attitude must reach a target state at a prescribed terminal time T_f dictated by communication windows or multi-vehicle coordination [1–3]. Meanwhile, these missions also require the attitude to converge to within a prescribed accuracy imposed by payload pointing requirements [4–6]. For spacecraft actuated by reaction wheels, fulfilling these requirements is further complicated by dual saturation, namely, torque saturation and momentum saturation. Stringent terminal-time requirements drive high torques that accelerate momentum accumulation. Once momentum saturation occurs, the spacecraft risks losing control authority.

Conventional finite-time and fixed-time control methods [7–12] cannot prescribe T_f directly. Prescribed-time control (PTC) [13, 14] addresses this through time-dependent scaling functions [15, 16] or time-varying high-gain feedback [17–19]. However, existing PTC strategies remain conservative; for instance, in Ref. [3], convergence occurs at 250 s for $T_f = 500$ s. This premature settling forces an idle phase and the aggressive initial effort accelerates momentum accumulation.

Beyond conservatism, the physical realizability of PTC under dual saturation has received little attention and entails two consequences. First, although torque saturation has been considered in certain PTC strategies [20–22], angular-momentum saturation is seldom treated as a coupled constraint; reaching the momentum limit causes complete loss of control authority regardless of whether torque commands remain within bounds. Second, T_f is typically treated as a free parameter without accounting for the minimum feasible maneuver time $T_{f,\min}$ imposed by dual saturation. Without $T_f \geq T_{f,\min}$, prescribed-time convergence is physically unrealizable, rendering the notion of prescribed time meaningless under practical constraints.

Prior constrained attitude control methods [23–25] provide torque-saturation guarantees but do not address angular-momentum limits. Optimization-based planning [26–29] is inherently open-loop, and model predictive control [30, 31] ensures only asymptotic convergence without angular-momentum constraints. No existing approach simultaneously

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achieves exact terminal-time convergence, prescribed accuracy, and dual-saturation compliance. A systematic solution remains to be developed.

Motivated by the above analysis, this note proposes an exact prescribed-time prescribed-accuracy control (EPTPAC) framework with a two-loop architecture. The outer loop generates a dual-saturation-feasible reference trajectory reaching the target precisely at T_f via constrained optimal control. The inner loop employs a nonsingular prescribed-time controller that drives tracking errors to within a user-specified accuracy before T_f , despite inertia uncertainties and disturbances. A parameter-synthesis methodology coordinates the two loops, ensuring closed-loop feasibility under dual saturation.

The main contributions of this note are as follows:

- 1) Exact terminal-time convergence with prescribed accuracy is achieved without premature settling.
- 2) Rigorous axis-by-axis feasibility conditions are derived to guarantee strict compliance with dual-saturation limits. Moreover, a minimum feasible maneuver time $T_{f,\min}$ is quantified to prevent physically unachievable prescriptions.
- 3) A modular two-loop architecture that decouples constraint enforcement from tracking performance is established, coordinated by a parameter-synthesis methodology requiring only three free designer choices.

II. Problem Description and Preliminaries

A. Problem Description

Let $\mathbb{R}$ denote the set of real numbers. $\|\cdot\|$ denotes the 2-norm of a vector or matrix, and $\mathbf{I}_n$ denotes the $n \times n$ identity matrix. $\lambda_{\min}(\mathbf{A})$ and $\lambda_{\max}(\mathbf{A})$ denote the extreme eigenvalues of $\mathbf{A}$. For $\mathbf{a} = [a_1, a_2, a_3]^\top \in \mathbb{R}^3$, $\mathbf{a}^\times$ denotes the skew-symmetric matrix $\mathbf{a}^\times \triangleq [0, -a_3, a_2; a_3, 0, -a_1; -a_2, a_1, 0]$.

The attitude kinematics and dynamics of a rigid spacecraft actuated by reaction wheels are governed by

$$\dot{\boldsymbol{\sigma}} = \mathbf{G}(\boldsymbol{\sigma})\boldsymbol{\omega}, \quad \mathbf{G}(\boldsymbol{\sigma}) = \frac{1}{4} \left[(1 - \|\boldsymbol{\sigma}\|^2)\mathbf{I}_3 + 2\boldsymbol{\sigma}^\times + 2\boldsymbol{\sigma}\boldsymbol{\sigma}^\top \right] \quad (1)$$

$$(\mathbf{J}_0 + \Delta\mathbf{J})\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega}^\times[(\mathbf{J}_0 + \Delta\mathbf{J})\boldsymbol{\omega} + \mathbf{h}_w] + \mathbf{d} + \boldsymbol{\tau}, \quad \boldsymbol{\tau} = -\dot{\mathbf{h}}_w \quad (2)$$

where $\boldsymbol{\sigma} \in \mathbb{R}^3$ (MRPs) and $\boldsymbol{\omega} \in \mathbb{R}^3$ denote the attitude and body angular velocity, $\mathbf{J}_0 \in \mathbb{R}^{3 \times 3}$ is the nominal inertia with uncertainty $\Delta\mathbf{J}$, $\mathbf{h}_w \in \mathbb{R}^3$ and $\boldsymbol{\tau} \in \mathbb{R}^3$ are the wheel angular momentum and control torque, and $\mathbf{d} \in \mathbb{R}^3$ is the external disturbance.

A three-axis wheel cluster aligned with the body axes is subject to dual-saturation constraints

$$|\tau_i(t)| \leq \tau_{i,\max}, \quad |h_{w,i}(t)| \leq h_{w,i,\max} \quad (3)$$

for each axis $i \in \{1, 2, 3\}$.

Let σ_d, ω_d denote the desired attitude and angular velocity. The tracking errors σ_e, ω_e are defined as

$$\sigma_e = \frac{(1 - \|\sigma_d\|^2)\sigma - (1 - \|\sigma\|^2)\sigma_d + 2\sigma_d^\times \sigma}{1 + \|\sigma\|^2\|\sigma_d\|^2 + 2\sigma^\top \sigma_d}, \quad \omega_e = \omega - \mathbf{R}(\sigma_e)\omega_d \quad (4)$$

where $\mathbf{R}(\sigma_e) = \mathbf{I}_3 + [8\sigma_e^\times \sigma_e^\times - 4(1 - \sigma_e^\top \sigma_e)\sigma_e^\times]/(1 + \sigma_e^\top \sigma_e)^2$ is the direction cosine matrix.

The error kinematics and dynamics take the form

$$\dot{\sigma}_e = \mathbf{G}(\sigma_e)\omega_e, \quad \mathbf{G}(\sigma_e) = \frac{1}{4} [(1 - \|\sigma_e\|^2)\mathbf{I}_3 + 2\sigma_e^\times + 2\sigma_e \sigma_e^\top] \quad (5)$$

$$\mathbf{J}_0 \dot{\omega}_e = \tau + \mathbf{T}_d + \mathbf{f} \quad (6)$$

where $\mathbf{f} = -\omega^\times(\mathbf{J}_0\omega + \mathbf{h}_w) + \mathbf{J}_0\omega_e^\times \mathbf{R}(\sigma_e)\omega_d - \mathbf{J}_0\mathbf{R}(\sigma_e)\dot{\omega}_d$ is the computable feedforward term, and $\mathbf{T}_d = -\Delta\mathbf{J}\dot{\omega}_e - \omega^\times\Delta\mathbf{J}\omega + \Delta\mathbf{J}(\omega_e^\times \mathbf{R}(\sigma_e)\omega_d - \mathbf{R}(\sigma_e)\dot{\omega}_d) + \mathbf{d}$ is the lumped perturbation.

The following assumptions are adopted throughout this work.

Assumption 1. *The prescribed terminal time satisfies*

$$T_f \geq T_{f,\min} \quad (7)$$

where $T_{f,\min} > 0$ is the minimum feasible maneuver time under the dual-saturation constraints (3). A practical estimate of $T_{f,\min}$ follows from Appendix A.

Assumption 2. *The lumped perturbation satisfies*

$$\|\mathbf{T}_d(t)\| \leq T_{d,\max}, \quad \forall t \in [0, T_f] \quad (8)$$

Assumption 3. *Let $\mathbf{H} = \mathbf{J}\omega + \mathbf{h}_w$ be the total angular momentum and $\mathbf{H}_0 = \mathbf{H}(0)$. The momentum drift satisfies*

$$\|\mathbf{H}(t) - \mathbf{H}_0\| \leq H_{\max}, \quad \forall t \in [0, T_f] \quad (9)$$

where $H_{\max} > 0$ is a known bound.

B. Preliminaries

Consider $\dot{\mathbf{x}} = \mathbf{g}(\mathbf{x}, t)$ with $\mathbf{g}(\mathbf{0}, t) \equiv \mathbf{0}$ and $\mathbf{g}$ locally Lipschitz in $\mathbf{x}$. The origin is prescribed-time stable if $\mathbf{x}(t) = \mathbf{0}$ for all $t \geq T_p$ [13], and prescribed-time prescribed-accuracy stable if $\|\mathbf{x}(t)\| \leq \varepsilon$ for all $t \geq T_p$ [32].

Lemma 1 ([33]). *Suppose there exists a Lyapunov function $V(\mathbf{x})$ such that*

$$\dot{V} \leq -\frac{\pi}{\eta T_p} \left(V^{1-\frac{\eta}{2}} + V^{1+\frac{\eta}{2}} \right) \quad (10)$$

where $0 < \eta < 1$ and $T_p > 0$. Then the origin is prescribed-time stable.

Theorem 1. *Suppose there exists a Lyapunov function $V(\mathbf{x})$ such that*

$$\dot{V} \leq \delta V^{\frac{1}{2}} - \frac{\pi}{\eta T_p} \left[(1 + \alpha) V^{1-\frac{\eta}{2}} + V^{1+\frac{\eta}{2}} \right] \quad (11)$$

where $\delta > 0$, $\alpha > 0$, $0 < \eta < 1$, and $T_p > 0$. Then the trajectory enters $V \leq V_\delta^$ with $V_\delta^* = \left(\frac{\delta \eta T_p}{\pi \alpha} \right)^{\frac{2}{1-\eta}}$ within T_p and remains therein.*

Proof. Let $K = \frac{\pi}{\eta T_p}$. From (11), $\dot{V} \leq \delta V^{1/2} - K[(1+\alpha)V^{1-\frac{\eta}{2}} + V^{1+\frac{\eta}{2}}]$. At $V = V_\delta^*$, $\delta V^{1/2} = K\alpha V^{1-\eta/2}$ holds by definition. For all $V \geq V_\delta^*$, since $V^{-(1-\eta)/2}$ is monotonically decreasing, $\delta V^{1/2} \leq K\alpha V^{1-\eta/2}$, yielding $\dot{V} \leq -K(V^{1-\frac{\eta}{2}} + V^{1+\frac{\eta}{2}})$ for all $V \geq V_\delta^*$. By Lemma 1, any trajectory with $V(0) > V_\delta^*$ enters $\{V \leq V_\delta^*\}$ within T_p . At $V = V_\delta^*$, $\dot{V} \leq -K(V^{1-\frac{\eta}{2}} + V^{1+\frac{\eta}{2}})$ ensures forward invariance. $\square$

III. Main Results

A. Outer-Loop Trajectory Planning Under Dual Saturation

A reference trajectory $(\sigma_d(t), \omega_d(t))$ is generated offline by solving the following constrained OCP over $[0, T_f]$, which embeds the kinematics, dynamics, and dual-saturation constraints:

$$\begin{aligned} \min_{\tau_{\text{ref}}(\cdot)} \quad & \mathcal{J} = \int_0^{T_f} \left(\|\tau_{\text{ref}}(t)\|^2 + \lambda \|\dot{\tau}_{\text{ref}}(t)\|^2 \right) dt \\ \text{s.t.} \quad & \dot{\sigma}(t) = \mathbf{G}(\sigma(t))\omega(t), \\ & \mathbf{J}_0 \dot{\omega}(t) + \omega(t)^\times (\mathbf{J}_0 \omega(t) + \mathbf{h}_w(t)) = \tau_{\text{ref}}(t), \\ & \dot{\mathbf{h}}_w(t) = -\tau_{\text{ref}}(t), \quad \mathbf{h}_w(0) = \mathbf{h}_{w,0}, \\ & \sigma(0) = \sigma_0, \quad \omega(0) = \omega_0, \\ & \sigma(T_f) = \sigma_{\text{target}}, \quad \omega(T_f) = \omega_{\text{target}}, \\ & |\tau_{\text{ref},i}(t)| \leq (1 - \gamma_u) \tau_{i,\max}, \quad i \in \{1, 2, 3\}, \\ & |h_{w,i}(t)| \leq (1 - \gamma_h) h_{w,i,\max}, \quad i \in \{1, 2, 3\} \end{aligned} \quad (12)$$

where $\lambda > 0$ balances control effort against smoothness, and $\gamma_u, \gamma_h \in (0, 1)$ are safety margins reserved for the inner loop. The OCP is solved via GPOPS-II [34] with piecewise-linear interpolation, yielding a reference trajectory with bounded angular velocity and acceleration: there exist constants $\omega_{d,\max} > 0$ and $\dot{\omega}_{d,\max} > 0$ such that $\|\omega_d(t)\| \leq \omega_{d,\max}$ and $\|\dot{\omega}_d(t)\| \leq \dot{\omega}_{d,\max}$ for all $t \in [0, T_f]$.

This decoupling of constrained planning from tracking control enables exact terminal-time convergence and dual-saturation compliance without the conservatism of existing PTC methods.

B. Inner-Loop Nonsingular Prescribed-Time Prescribed-Accuracy Tracking Control

Define the sliding mode surface

$$s = \omega_e + \mathbf{Q}(\sigma_e) \quad (13)$$

with the nonlinear coupling term $\mathbf{Q}(\sigma_e)$ defined as

$$\mathbf{Q}(\sigma_e) = c_1 k_1 \frac{\sigma_e}{(\|\sigma_e\| + \varepsilon_1)^\eta} + c_1 k_2 \frac{\sigma_e}{(\|\sigma_e\| + \varepsilon_1)^{-\eta}} \quad (14)$$

where $0 < \eta < 1$, $\varepsilon_1 > 0$, $T_{p1} > 0$, and $\alpha_1 > 0$ are design parameters. The gains are given by

$$c_1 = \frac{\pi}{\eta T_{p1}}, \quad k_1 = (1 + \alpha_1) 2^{1+\frac{3}{2}\eta}, \quad k_2 = 2^{1-\frac{\eta}{2}}$$

Theorem 2. *Consider the attitude error kinematics (5) with the sliding mode surface (13), and suppose that the sliding mode variable $s(t)$ remains within the bounded set $\|s(t)\| \leq s_{\max}$. Then the attitude error $\sigma_e(t)$ converges to the bounded set*

$$\|\sigma_e(t)\| \leq \max \{\varepsilon_{\delta_1}, \varepsilon_1\} \quad (15)$$

within T_{p1} and remains therein for all $t \geq T_{p1}$, where $\varepsilon_{\delta_1} = \sqrt{2} (\delta_1 \eta T_{p1} / (\pi \alpha_1))^{1/(1-\eta)}$ and $\delta_1 = \sqrt{2} s_{\max} / 2$.

Proof. Define the Lyapunov function candidate

$$V_1 = \frac{1}{2} \|\sigma_e\|^2 \quad (16)$$

Differentiating along (5) and substituting $\omega_e = s - \mathbf{Q}(\sigma_e)$ yields

$$\dot{V}_1 \leq \|\mathbf{G}(\sigma_e)\| \|\sigma_e\| s_{\max} - \sigma_e^\top \mathbf{G}(\sigma_e) \mathbf{Q}(\sigma_e) \quad (17)$$

For $\|\sigma_e\| \geq \varepsilon_1$, invoking $\|G(\sigma_e)\| < \frac{1}{2}$ (which holds for $\|\sigma_e\| < 1$) and the inequality $\|x\|^2/(\|x\| + \varepsilon)^\eta \geq \frac{1}{2\eta}\|x\|^{2-\eta}$,

$$\begin{aligned}\dot{V}_1 &\leq \frac{1}{2}\|\sigma_e\|s_{\max} - \frac{c_1}{4}\left[k_1\frac{\|\sigma_e\|^2}{(\|\sigma_e\| + \varepsilon_1)^\eta} + k_2\frac{\|\sigma_e\|^2}{(\|\sigma_e\| + \varepsilon_1)^{-\eta}}\right] \\ &\leq \frac{\sqrt{2}}{2}V_1^{1/2}s_{\max} - \frac{c_1}{4}\left[k_12^{-\eta}(2V_1)^{1-\eta/2} + k_2(2V_1)^{1+\eta/2}\right]\end{aligned}\quad (18)$$

Substituting the gains c_1, k_1, k_2 and simplifying,

$$\dot{V}_1 \leq \delta_1 V_1^{1/2} - \frac{\pi}{\eta T_{p1}} \left[(1 + \alpha_1) V_1^{1-\frac{\eta}{2}} + V_1^{1+\frac{\eta}{2}} \right] \quad (19)$$

This satisfies the condition of Theorem 1. Hence V_1 enters $\{V_1 < V_{\delta_1}^*\}$ before T_{p1} with $V_{\delta_1}^* = (\frac{\delta_1 \eta T_{p1}}{\pi \alpha_1})^{\frac{2}{1-\eta}}$. Converting back, $\|\sigma_e(t)\| \leq \sqrt{2V_{\delta_1}^*} = \varepsilon_{\delta_1}$. The regularized set $\{\|\sigma_e\| \leq \varepsilon_1\}$ is positively invariant. Thus $\|\sigma_e(t)\| \leq \max\{\varepsilon_{\delta_1}, \varepsilon_1\}$ for all $t \geq T_{p1}$. $\square$

The inner-loop torque command is designed as

$$\tau = -f(t) - J_0 \dot{Q}(\sigma_e) + \tau_c \quad (20)$$

with the prescribed-time stabilizing term

$$\tau_c = -c_2 \left[k_3 \frac{s}{(\|s\| + \varepsilon_2)^\eta} + k_4 \frac{s}{(\|s\| + \varepsilon_2)^{-\eta}} \right] \quad (21)$$

and gains

$$c_2 = \frac{\pi}{\eta T_{p2}}, \quad k_3 = (1 + \alpha_2) 2^{-1+\frac{3}{2}\eta} \lambda_{\max}(J_0)^{1-\frac{\eta}{2}}, \quad k_4 = 2^{-1-\frac{\eta}{2}} \lambda_{\max}(J_0)^{1+\frac{\eta}{2}} \quad (22)$$

where $T_{p2} > 0$ and $\alpha_2 > 0$ are design parameters. Substituting (20) into (6) and noting $\dot{\omega}_e = \dot{s} - \dot{Q}(\sigma_e)$ gives the closed-loop dynamics

$$J_0 \dot{s} = \tau_c + T_d(t) \quad (23)$$

Theorem 3. Consider the spacecraft tracking error system described by (5) and (6), and suppose that Assumption 2 is satisfied. If the control law (20) is applied, then the sliding mode variable $s(t)$ converges to the bounded set

$$\|s(t)\| \leq \max\{\varepsilon_{\delta_2}, \varepsilon_2\} \quad (24)$$

within T_{p2} and remains therein for all $t \geq T_{p2}$, where $\varepsilon_{\delta_2} = \sqrt{2/\lambda_{\min}(J_0)} (\delta_2 \eta T_{p2}/(\pi \alpha_2))^{1/(1-\eta)}$ and $\delta_2 = T_{d,\max} \sqrt{2/\lambda_{\min}(J_0)}$.

Proof. Define the Lyapunov function candidate

$$V_2 = \frac{1}{2} \mathbf{s}^\top \mathbf{J}_0 \mathbf{s} \quad (25)$$

Differentiating along (23) and using $\|\mathbf{s}\| \leq \sqrt{2V_2/\lambda_{\min}(\mathbf{J}_0)}$,

$$\dot{V}_2 = \mathbf{s}^\top \boldsymbol{\tau}_c + \mathbf{s}^\top \mathbf{T}_d \leq \mathbf{s}^\top \boldsymbol{\tau}_c + \delta_2 V_2^{1/2} \quad (26)$$

with $\delta_2 = T_{d,\max} \sqrt{2/\lambda_{\min}(\mathbf{J}_0)}$. For $\|\mathbf{s}\| \geq \varepsilon_2$, applying the same norm inequality as in the proof of Theorem 2 and substituting the gains (22),

$$\begin{aligned} \mathbf{s}^\top \boldsymbol{\tau}_c &\leq -c_2 \left[k_3 \frac{\|\mathbf{s}\|^2}{(\|\mathbf{s}\| + \varepsilon_2)^\eta} + k_4 \frac{\|\mathbf{s}\|^2}{(\|\mathbf{s}\| + \varepsilon_2)^{-\eta}} \right] \\ &\leq -c_2 \left[k_3 2^{-\eta} \|\mathbf{s}\|^{2-\eta} + k_4 \|\mathbf{s}\|^{2+\eta} \right] \\ &\leq -\frac{\pi}{\eta T_{p2}} \left[(1 + \alpha_2) V_2^{1-\eta/2} + V_2^{1+\eta/2} \right] \end{aligned} \quad (27)$$

Substituting this bound into the expression for $\dot{V}_2$,

$$\dot{V}_2 \leq \delta_2 V_2^{1/2} - \frac{\pi}{\eta T_{p2}} \left[(1 + \alpha_2) V_2^{1-\eta/2} + V_2^{1+\eta/2} \right] \quad (28)$$

This satisfies the condition of Theorem 1. Hence V_2 enters $\{V_2 \leq V_{\delta_2}^*\}$ within T_{p2} with $V_{\delta_2}^* = (\frac{\delta_2 \eta T_{p2}}{\pi \alpha_2})^{\frac{2}{1-\eta}}$. Converting back, $\|\mathbf{s}(t)\| \leq \sqrt{2V_{\delta_2}^*/\lambda_{\min}(\mathbf{J}_0)} = \varepsilon_{\delta_2}$. The set $\{\|\mathbf{s}\| \leq \varepsilon_2\}$ is positively invariant. Thus $\|\mathbf{s}(t)\| \leq \max\{\varepsilon_{\delta_2}, \varepsilon_2\}$ for all $t \geq T_{p2}$. $\square$

By appropriately selecting $\alpha_1, \alpha_2 > 0$, the designer can ensure $\varepsilon_{\delta_1} < \varepsilon_1$ and $\varepsilon_{\delta_2} < \varepsilon_2$, so that the user-specified tolerances $\varepsilon_1, \varepsilon_2$ govern the ultimate convergence bounds. Theorems 2 and 3 jointly ensure that the tracking errors converge in two stages. First, $\|\mathbf{s}(t)\| \leq \max\{\varepsilon_{\delta_2}, \varepsilon_2\}$ for $t \geq T_{p2}$. Subsequently, $\|\boldsymbol{\sigma}_e(t)\| \leq \max\{\varepsilon_{\delta_1}, \varepsilon_1\}$ for $t \geq T_{p1} + T_{p2}$. Both bounds remain valid thereafter. Setting $T_{p1} + T_{p2} < T_f$ guarantees settling before the terminal time.

The following theorems verify dual-saturation compliance under the condition

$$\|\boldsymbol{\sigma}_e(t)\| \leq \varepsilon_1, \quad \|\mathbf{s}(t)\| \leq \varepsilon_2 \quad (29)$$

Theorem 4. Consider the inner-loop control law (20). Under the condition (29), if the safety margin γ_u and

prescribed-accuracy set $(\varepsilon_1, \varepsilon_2)$ are selected such that

$$\Delta_{f,i,\max}(\varepsilon_1, \varepsilon_2) + \overline{(J_0 \dot{Q})}_i(\varepsilon_1, \varepsilon_2) + \bar{\tau}_c(\varepsilon_2) \leq \gamma_u \tau_{i,\max}, \quad i = 1, 2, 3 \quad (30)$$

where $\Delta_{f,i,\max}(\varepsilon_1, \varepsilon_2)$, $\overline{(J_0 \dot{Q})}_i(\varepsilon_1, \varepsilon_2)$, and $\bar{\tau}_c(\varepsilon_2)$ are explicit bounds derived in the proof, then the actuator torques satisfy $|\tau_i(t)| \leq \tau_{i,\max}$ for all $i = 1, 2, 3$.

Proof. From the decomposition $-\mathbf{f}(t) = \boldsymbol{\tau}_{\text{ref}}(t) + \boldsymbol{\Delta}_f(t)$ (see Appendix B), substituting (20) and applying the triangle inequality yields, for each axis $i \in \{1, 2, 3\}$,

$$|\tau_i| \leq |\tau_{\text{ref},i}| + |\Delta_{f,i}| + |(\mathbf{J}_0 \dot{\mathbf{Q}})_i| + |\tau_{c,i}| \quad (31)$$

The outer loop ensures $|\tau_{\text{ref},i}(t)| \leq (1 - \gamma_u) \tau_{i,\max}$ for all $t \in [0, T_f]$. Under condition (29), the angular velocity error satisfies

$$\|\boldsymbol{\omega}_e(t)\| \leq \varepsilon_2 + \bar{Q} \quad (32)$$

where

$$\bar{Q} = \sup_{\|\boldsymbol{\sigma}_e\| \leq \varepsilon_1} \|\mathbf{Q}(\boldsymbol{\sigma}_e)\| \leq c_1 (k_1 2^{-\eta} \varepsilon_1^{1-\eta} + k_2 2^\eta \varepsilon_1^{1+\eta}) \quad (33)$$

(i) *Bound on $|\Delta_{f,i}|$.* From Appendix B and applying the bounds (33) and the reference bounds from Section III, for each $i = 1, 2, 3$,

$$\begin{aligned} |\Delta_{f,i}(t)| \leq \Delta_{f,i,\max}(\varepsilon_1, \varepsilon_2) &= (\varepsilon_2 + \bar{Q} + \omega_{d,\max}) H_{\max} + (\varepsilon_2 + \bar{Q}) H_0 + 4\varepsilon_1 \omega_{d,\max} H_0 + 2\omega_{d,\max} H_0 \\ &\quad + \hat{\nu}_i (\varepsilon_2 + \bar{Q} + \omega_{d,\max})^2 + \mu_i (\varepsilon_2 + \bar{Q}) \omega_{d,\max} + 4\mu_i \varepsilon_1 \dot{\omega}_{d,\max} \end{aligned} \quad (34)$$

where $\mu_i = \sum_{j=1}^3 |(J_0)_{ij}|$, $\nu_i = \sum_{j=1}^3 |(\Delta J)_{ij}|$, $\hat{\nu}_i = \sum_{j \neq i} \nu_j$, $H_0 = \|\mathbf{H}_0\|$, and $H_{\max}$ is from Assumption 3.

(ii) *Bound on $|\tau_{c,i}|$.* Since $\|\mathbf{s}(t)\| \leq \varepsilon_2$,

$$|\tau_{c,i}(t)| \leq \bar{\tau}_c(\varepsilon_2) = c_2 (k_3 2^{-\eta} \varepsilon_2^{1-\eta} + k_4 2^\eta \varepsilon_2^{1+\eta}) \quad (35)$$

(iii) *Bound on $|(\mathbf{J}_0 \dot{\mathbf{Q}})_i|$.* From $\dot{\mathbf{Q}} = \frac{\partial \mathbf{Q}}{\partial \boldsymbol{\sigma}_e} \mathbf{G}(\boldsymbol{\sigma}_e) \boldsymbol{\omega}_e$ and $\|\mathbf{G}(\boldsymbol{\sigma}_e)\| < 1/2$,

$$\|\dot{\mathbf{Q}}(t)\| \leq \left\| \frac{\partial \mathbf{Q}}{\partial \boldsymbol{\sigma}_e} \right\| \frac{1 + \varepsilon_1^2}{4} (\varepsilon_2 + \bar{Q}) \quad (36)$$

For $\|\sigma_e\| \leq \varepsilon_1$, the Jacobian norm is bounded by

$$\left\| \frac{\partial \mathbf{Q}}{\partial \sigma_e} \right\| \leq c_1 \left[k_1 (2\varepsilon_1)^{-\eta} + k_2 (2\varepsilon_1)^\eta + \varepsilon_1 (k_1 \eta (2\varepsilon_1)^{-\eta-1} + k_2 \eta (2\varepsilon_1)^{\eta-1}) \right] \triangleq \Psi(\varepsilon_1) \quad (37)$$

Hence

$$|(J_0 \dot{\mathbf{Q}})_i| \leq \mu_i \Psi(\varepsilon_1) \frac{1 + \varepsilon_1^2}{4} (\varepsilon_2 + \bar{Q}) \triangleq \overline{(J_0 \dot{\mathbf{Q}})_i}(\varepsilon_1, \varepsilon_2) \quad (38)$$

Finally, substituting the bounds (i)–(iii) into (31), for each axis i , the control torque satisfies

$$|\tau_i(t)| \leq (1 - \gamma_u) \tau_{i,\max} + \Delta_{f,i,\max}(\varepsilon_1, \varepsilon_2) + \overline{(J_0 \dot{\mathbf{Q}})_i}(\varepsilon_1, \varepsilon_2) + \bar{\tau}_c(\varepsilon_2)$$

Consequently, provided that condition (30) is satisfied, the control torque satisfies $|\tau_i(t)| \leq \tau_{i,\max}$ for all $i \in \{1, 2, 3\}$. $\square$

Theorem 5. *Consider the inner-loop control law (20). Under the condition (29), if the safety margin γ_h and prescribed-accuracy set $(\varepsilon_1, \varepsilon_2)$ are selected such that*

$$H_{\max} + 2H_0 + 4\mu_i \varepsilon_1 \omega_{d,\max} + (\mu_i + \nu_i)(\varepsilon_2 + \bar{Q}) + \nu_i \omega_{d,\max} \leq \gamma_h h_{w,i,\max}, \quad i = 1, 2, 3 \quad (39)$$

where $\bar{Q}$ is defined in (33), $\mu_i = \sum_{j=1}^3 |(J_0)_{ij}|$, and $\nu_i = \sum_{j=1}^3 |(\Delta J)_{ij}|$, then the wheel angular momentum $|h_{w,i}(t)| \leq h_{w,i,\max}$ holds for all $i = 1, 2, 3$.

Proof. The wheel momentum $\mathbf{h}_w = \mathbf{H} - (\mathbf{J}_0 + \Delta \mathbf{J})\boldsymbol{\omega}$ can be related to the reference momentum $\mathbf{h}_{w,d} = \mathbf{H}_d(t) - \mathbf{J}_0 \boldsymbol{\omega}_d$ as follows. Using $\boldsymbol{\omega} = \boldsymbol{\omega}_e + \mathbf{R}(\sigma_e)\boldsymbol{\omega}_d$ and writing $\mathbf{H} = \mathbf{H}_0 + (\mathbf{H} - \mathbf{H}_0)$ and $\mathbf{H}_d = \mathbf{H}_0 + (\mathbf{H}_d - \mathbf{H}_0)$:

$$\mathbf{h}_w = \mathbf{h}_{w,d} + (\mathbf{H} - \mathbf{H}_0) + (\mathbf{H}_0 - \mathbf{H}_d) - \mathbf{J}_0(\mathbf{R}(\sigma_e) - \mathbf{I})\boldsymbol{\omega}_d - (\mathbf{J}_0 + \Delta \mathbf{J})\boldsymbol{\omega}_e - \Delta \mathbf{J} \mathbf{R}(\sigma_e)\boldsymbol{\omega}_d$$

By the condition (29), $\|\boldsymbol{\omega}_e\| \leq \varepsilon_2 + \bar{Q}$. Together with the reference bounds $\|\boldsymbol{\omega}_d\| \leq \omega_{d,\max}$ from Section III, $\|\mathbf{H} - \mathbf{H}_0\| < H_{\max}$ from Assumption 3, $\|\mathbf{H}_0 - \mathbf{H}_d(t)\| \leq 2H_0$ (see Appendix B), $\|\mathbf{I} - \mathbf{R}(\sigma_e)\| \leq 4\|\sigma_e\| \leq 4\varepsilon_1$, and the outer-loop constraint $|h_{w,d,i}| \leq (1 - \gamma_h)h_{w,i,\max}$, applying the triangle inequality to each axis i yields

$$\begin{aligned} |h_{w,i}| &\leq |h_{w,d,i}| + H_{\max} + 2H_0 + 4\mu_i \varepsilon_1 \omega_{d,\max} + (\mu_i + \nu_i)\|\boldsymbol{\omega}_e\| + \nu_i \|\boldsymbol{\omega}_d\| \\ &\leq |h_{w,d,i}| + H_{\max} + 2H_0 + 4\mu_i \varepsilon_1 \omega_{d,\max} + (\mu_i + \nu_i)(\varepsilon_2 + \bar{Q}) + \nu_i \omega_{d,\max} \\ &\leq (1 - \gamma_h)h_{w,i,\max} + H_{\max} + 2H_0 + 4\mu_i \varepsilon_1 \omega_{d,\max} + (\mu_i + \nu_i)(\varepsilon_2 + \bar{Q}) + \nu_i \omega_{d,\max} \end{aligned} \quad (40)$$

Therefore, if (39) holds, the wheel angular momentum $|h_{w,i}(t)| \leq h_{w,i,\max}$ holds for all $i = 1, 2, 3$. $\square$

Remark 1. With $\sigma_d(0) = \sigma(0)$ and $\omega_d(0) = \omega(0)$, the initial errors are zero. By Theorems 2 and 3, the sets $\{\|s\| \leq \varepsilon_2\}$ and $\{\|\sigma_e\| \leq \varepsilon_1\}$ are forward invariant. Hence the conditions of Theorems 4 and 5 hold over $[0, T_f]$, ensuring dual-saturation compliance throughout the maneuver.

Remark 2. Within the prescribed accuracy sets, Δ_f and $\mathbf{J}_0 \dot{\omega}_e$ are small, so the dominant torque disturbance is $\mathbf{T}_d$. A practical guideline is $\gamma_u > T_{d,\max}/\tau_{i,\max}$ for each axis i .

Remark 3. Similarly, the term $(\mu_i + \nu_i)(\varepsilon_2 + \bar{Q})$ is negligible within the prescribed accuracy sets, so the momentum margin is governed by $H_{\max}$, $2H_0$, and $\nu_i \omega_{d,\max}$. A practical guideline is $\gamma_h > (H_{\max} + 2H_0 + \nu_i \omega_{d,\max})/h_{w,i,\max}$ for each axis i .

C. Parameter-Synthesis Methodology

Fig. 1 summarizes the parameter-synthesis procedure for the EPTPAC framework.

- 1) **Mission inputs.** Specify T_f , $\varepsilon_{\text{mission}}$, $\mathbf{J}_0$, $\tau_{\max}$, $h_{w,\max}$, and $T_{d,\max}$.
- 2) **Margins and feasibility.** Select γ_u, γ_h via Remarks 2–3; verify $T_f \geq T_{f,\min}$ via Appendix A.
- 3) **Inner-loop synthesis.** Set $\varepsilon_1 = \varepsilon_{\text{mission}}$, $\eta \in (0, 1)$, $T_{p1}, T_{p2} > 0$ with $T_{p1} + T_{p2} < T_f$, and select ε_2 via Remark 4.

With $s_{\max} = \varepsilon_2$, $\delta_1 = \frac{\sqrt{2}}{2}s_{\max}$, and $\delta_2 = T_{d,\max}\sqrt{2/\lambda_{\min}(\mathbf{J}_0)}$, the required gains satisfy

$$\alpha_1 \geq \frac{\delta_1 \eta T_{p1}}{\pi} \left(\frac{\sqrt{2}}{\varepsilon_1} \right)^{1-\eta}, \quad \alpha_2 \geq \frac{\delta_2 \eta T_{p2}}{\pi} \left(\frac{\sqrt{2}}{\varepsilon_2 \sqrt{\lambda_{\min}(\mathbf{J}_0)}} \right)^{1-\eta} \quad (41)$$

- 4) **Saturation verification.** Check conditions in Theorems 4–5; if violated, increase T_f , relax ε_1 , or increase γ_u, γ_h .

The framework requires only three free designer choices, namely η , T_{p1} , and T_{p2} , from the mission inputs. All remaining parameters are systematically derived via the analytical conditions provided in this study.

Remark 4. To avoid unbalanced gains, enforce $\alpha_1 = \alpha_2 = \kappa \alpha$ with $\kappa > 1$ by solving $\alpha_{1,\min} = \alpha_{2,\min}$ in (41). This yields

$$\varepsilon_2 = \left(\frac{2T_{d,\max} T_{p2}}{T_{p1} \lambda_{\min}(\mathbf{J}_0)^{1-\frac{\eta}{2}}} \right)^{\frac{1}{2-\eta}} \varepsilon_1^{\frac{1-\eta}{2-\eta}}, \quad \alpha = \frac{\delta_1 \eta T_{p1}}{\pi} \left(\frac{\sqrt{2}}{\varepsilon_1} \right)^{1-\eta} \quad (42)$$

followed by $\alpha_1 = \alpha_2 = \kappa \alpha$. This provides a unique parameter set from mission inputs.

IV. Simulation Results

Numerical simulations validate the proposed EPTPAC framework under dual saturation. Table 1 summarizes the spacecraft parameters, actuator limits, and boundary conditions. The simulation horizon extends to $T_{\text{sim}} > T_f$, with the reference held at the target for $t \geq T_f$.

The nominal trajectory reference (σ_d, ω_d) is generated offline by solving the constrained OCP (12) via GPOPS-II with $\lambda = 0.1$ and $N = 20$. The discrete nodal torques are converted to continuous-time commands via piecewise-linear

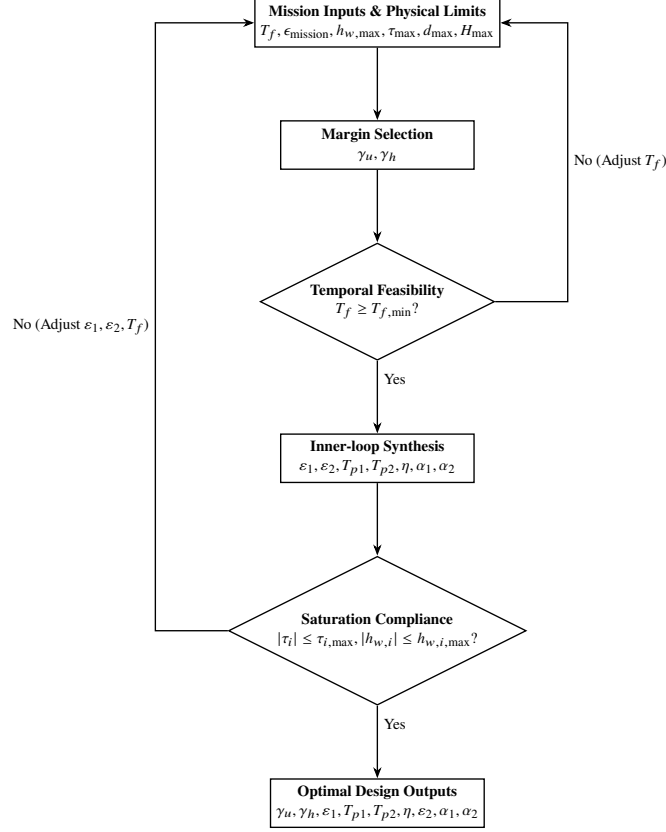


Fig. 1 Parameter-synthesis methodology.

interpolation. Following Remarks 2–3, $\gamma_u = 0.2$ and $\gamma_h = 0.1$ are adopted. The minimum feasible terminal time is $T_{f,\min} \approx 95$ s via Appendix A; all cases use $T_f \geq 110$ s to ensure feasibility.

Three cases are conducted to validate different aspects of the framework: Case 1 verifies fundamental performance with dual-saturation compliance; Case 2 examines the prescribed-time and prescribed-accuracy characteristics simultaneously; and Case 3 benchmarks the proposed framework against a representative torque-saturated prescribed-time controller from [21] (hereafter Ref [21]) to demonstrate the advantages of the proposed framework in achieving exact terminal-time convergence and dual-saturation feasibility.

A. Case 1: Nominal Performance Validation

The mission parameters are as follows: terminal time $T_f = 120$ s, mission accuracy $\varepsilon_1 = 10^{-5}$, prescribed-time exponent $\eta = 0.2$, and time allocations $T_{p2} = 15$ s, $T_{p1} = 100$ s. Following the synthesis methodology in Section III.C with $\kappa = 1.2$, the inner-loop parameters are determined to be $\varepsilon_2 = 6.43 \times 10^{-5}$ and $\alpha_1 = \alpha_2 = 6.34$. The practical convergence bounds $\varepsilon_{\delta_1} = 7.96 \times 10^{-6}$ and $\varepsilon_{\delta_2} = 5.13 \times 10^{-5}$ satisfy $\varepsilon_{\delta_i} < \varepsilon_i$.

The simulation results are illustrated in Figs. 2–4. Fig. 2 shows that the spacecraft attitude and angular velocity

Table 1 Common simulation settings and boundary conditions

Symbol	Value
$\mathbf{J}_0$	$\text{diag}([200, 250, 300]) \text{ kg} \cdot \text{m}^2$
$\Delta \mathbf{J}$	$5\% \mathbf{J}_0$
$\tau_{i,\max}$	$0.2 \text{ N} \cdot \text{m}$
$h_{w,i,\max}$	$4.0 \text{ kg} \cdot \text{m}^2/\text{s}$
$\mathbf{h}_{w,0}$	$[0, 0, 0]^\top \text{ kg} \cdot \text{m}^2/\text{s}$
$\mathbf{d}(t)$	$0.01 [\sin(2t), \cos(t), \cos(t+2)]^\top \text{ N} \cdot \text{m}$
$T_{d,\max}$	$0.03 \text{ N} \cdot \text{m}$
$H_{\max}$	$0.1 \text{ kg} \cdot \text{m}^2/\text{s}$
$\boldsymbol{\sigma}(0)$	$[0.2, 0.3, -0.3]^\top$
$\boldsymbol{\sigma}_{\text{target}}$	$[0, 0, 0]^\top$
$\boldsymbol{\omega}(0)$	$[0, 0, 0]^\top \text{ rad/s}$
$\boldsymbol{\omega}_{\text{target}}$	$[0, 0, 0]^\top \text{ rad/s}$

closely track the reference trajectory, which reaches the target state at $t = T_f$. As shown in Fig. 3, all tracking errors remain within the prescribed-accuracy sets throughout the maneuver. Fig. 4 confirms that both control torques and wheel momentum remain strictly within their respective limits over $[0, T_f]$.

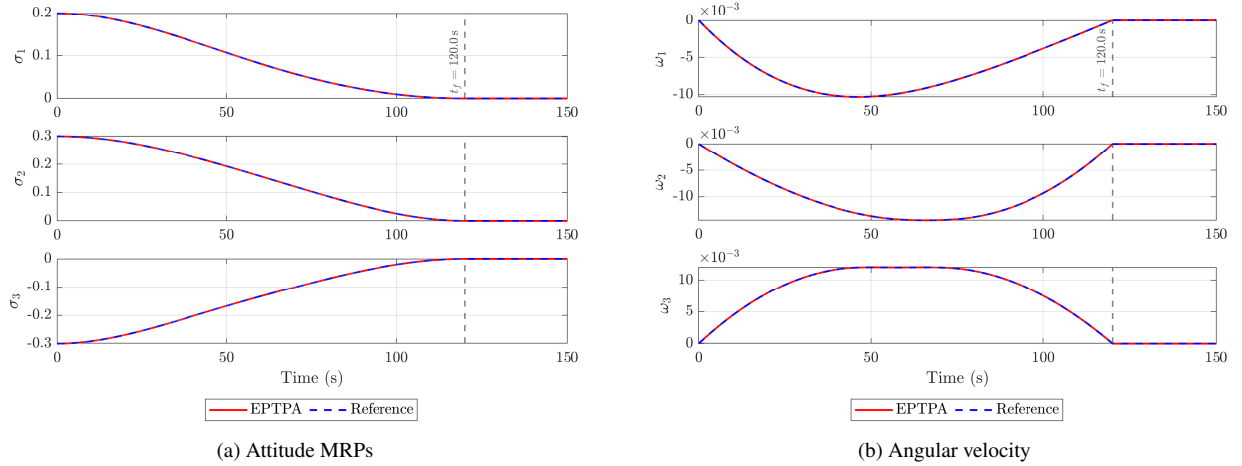


Fig. 2 Case 1: Reference tracking in MRPs and angular rates.

In summary, Case 1 validates the baseline performance of the EPTPAC framework under dual saturation.

B. Case 2: Prescribed Performance Assessment

This case evaluates the two defining characteristics of the EPTPAC framework, exact terminal-time convergence and prescribed-accuracy tunability, in a unified assessment.

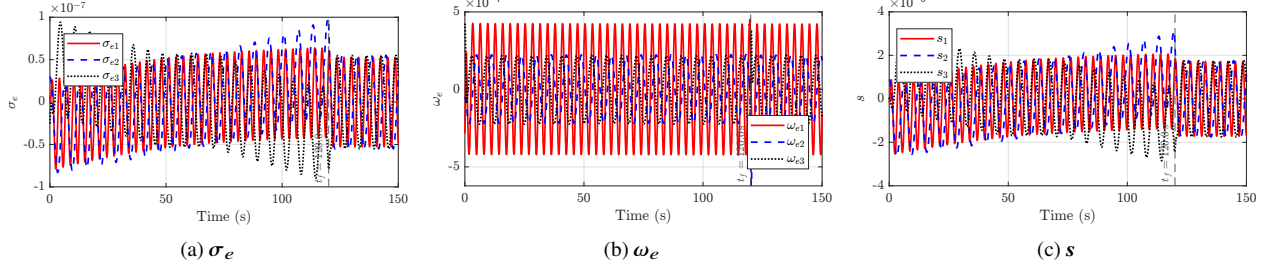


Fig. 3 Case 1: Error responses

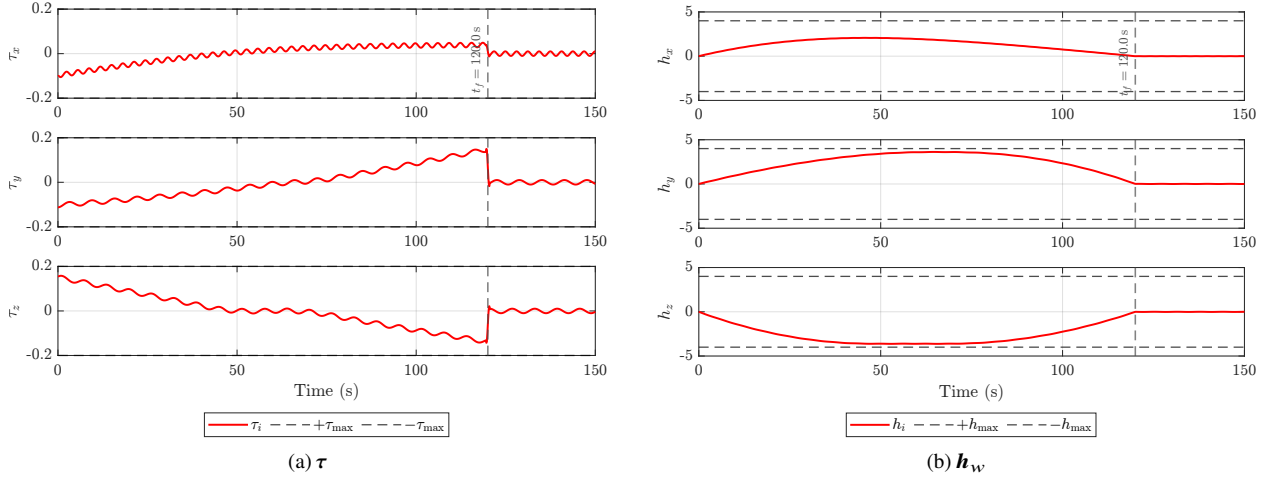


Fig. 4 Case 1: Torque and wheel momentum.

1. Time-flexibility.

The terminal time is swept over $T_f \in \{110, 120, 130, 140\}$ s, with $\varepsilon_1 = 10^{-5}$, $\eta = 0.2$, $T_{p2} = 15$ s, and $T_{p1} = T_f - 20$ s. For each T_f , the OCP generates a constraint-feasible reference trajectory, and the inner-loop parameters are synthesized via the balanced rule with $\kappa = 1.2$ following the procedure in Section III.C.

Fig. 5 shows that the target-pointing error converges at the respective terminal times for all tested values, confirming exact terminal-time convergence. Shorter T_f requires more aggressive control, yet the torque and momentum remain within limits for all T_f .

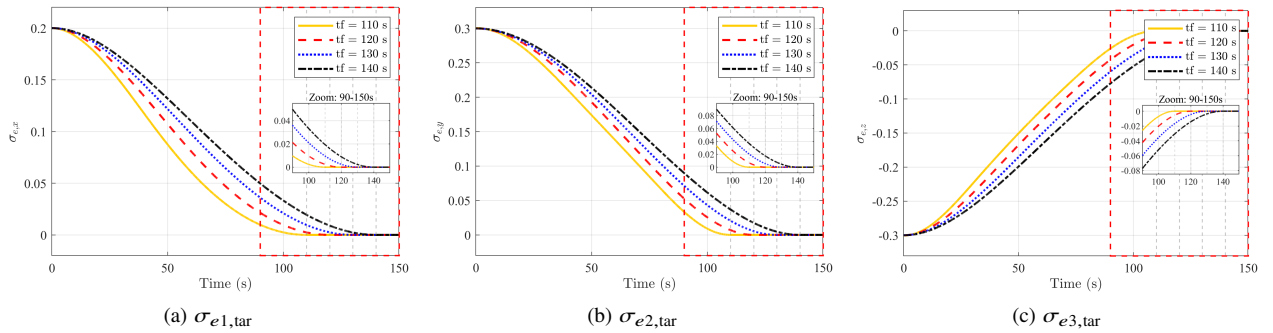


Fig. 5 Case 2: MRP error components for different T_f

The attitude tracking error enters the prescribed accuracy only 0.24–0.25 s prior to T_f , corresponding to a relative timing error below 0.06%.

2. Accuracy-sweep.

To evaluate accuracy tunability, the prescribed accuracy is swept over $\varepsilon_1 \in \{10^{-4}, 10^{-5}, 10^{-6}, 10^{-7}\}$ with $T_f = 120$ s fixed. For each ε_1 , the inner-loop parameters are synthesized via the balanced rule with $\kappa = 1.2$ following Section III.C.

As shown in Fig. 6, all four settings achieve their prescribed accuracies within T_f . Higher precision requires higher gains: tightening ε_1 from 10^{-4} to 10^{-7} increases $\alpha_1 = \alpha_2$ from 4.30 to 24.78.

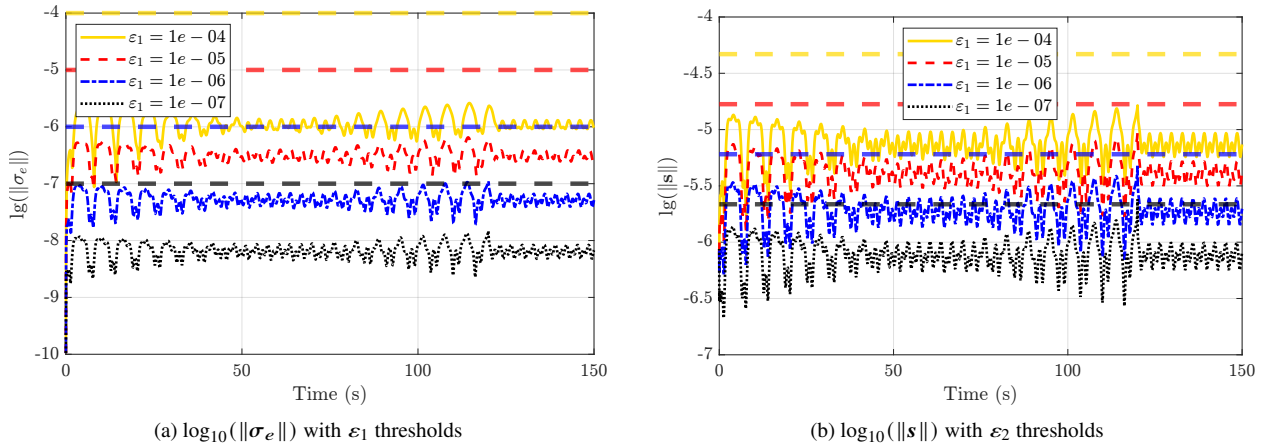


Fig. 6 Case 2: Log-scale comparison of $\|\sigma_e\|$ and $\|s\|$ for different ε_1

Case 2 confirms that the EPTPAC framework simultaneously delivers exact terminal-time convergence with sub-percent temporal precision and tunable steady-state accuracy across four orders of magnitude.

C. Case 3: Comparison with Ref. [21]

To benchmark the advantage of the proposed framework against existing methods, this case compares EPTPAC with the representative torque-saturated prescribed-time controller from Ref [21], which addresses torque limits but does not account for wheel-momentum constraints. Two scenarios are conducted: Case 3.1 considers torque saturation only, thereby providing a fair comparison under identical constraint assumptions; Case 3.2 enforces dual saturation to assess how the additional momentum constraint affects controller performance.

1. Case 3.1: Conservatism under torque-only saturation

This scenario compares terminal-time tracking performance under torque saturation ($\tau_{i,\max} = 0.2$ N · m) with $T_f = 300$ s.

As shown in Fig. 7, Ref. [21] exhibits pronounced conservatism, converging around 150 s despite the prescribed

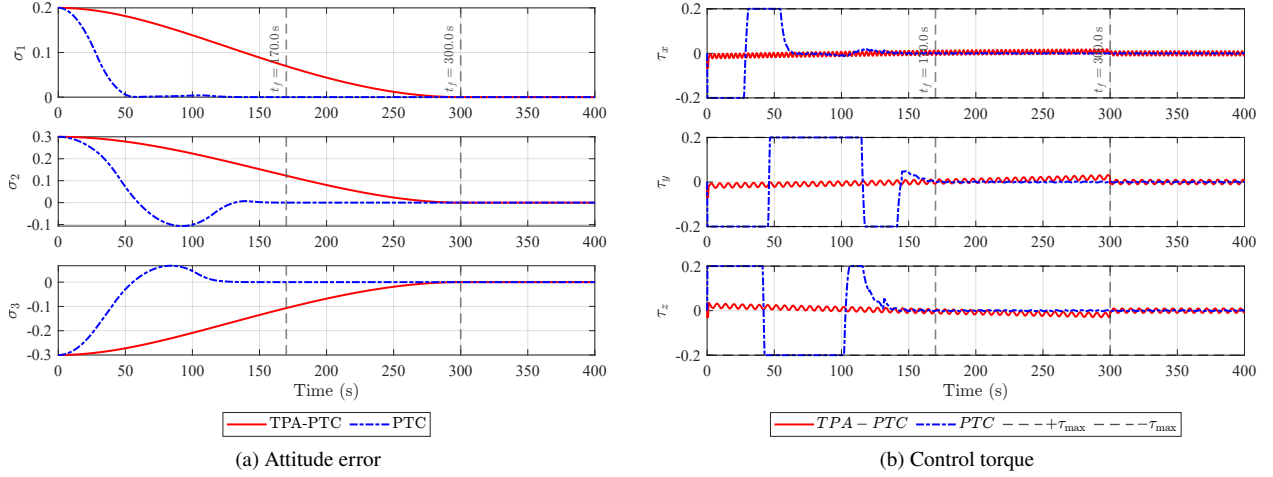


Fig. 7 Case 3.1: Performance comparison under torque-only saturation

$T_f = 300$ s. In contrast, EPTPAC achieves exact convergence at $t = 300$ s with smooth control effort.

2. Case 3.2: Loss of physical realizability under dual saturation

This scenario examines the consequences when physical constraints are violated. Under torque-only saturation with T_f reduced to 150 s, Ref. [21] loses prescribed-time convergence entirely, lagging until 230 s, because its high-gain mechanism does not account for the physical feasibility limit $T_{f,\min}$. Introducing the additional angular-momentum constraint ($h_{w,i,\max} = 4.0 \text{ kg} \cdot \text{m}^2/\text{s}$) with $T_f = 300$ s further exposes the vulnerability.

As shown in Fig. 8, Ref. [21] neglects angular-momentum constraints and commands impulsive torques, rapidly saturating the wheels. The resulting loss of control authority leads to uncontrolled drift and severely delayed convergence at 410 s, far exceeding the prescribed 300 s. In contrast, EPTPAC strategically modulates the torque profile to prevent momentum saturation, achieving exact convergence at $t = 300$ s.

In summary, Case 3 demonstrates that existing PTC methods suffer from pronounced conservatism (Case 3.1) and lose convergence guarantees when physical constraints are violated (Case 3.2). The EPTPAC framework circumvents both issues through constraint-aware co-design of the planning and control layers.

V. Conclusion

This note proposed a two-loop EPTPAC framework for spacecraft actuated by reaction wheels to achieve prescribed accuracy at a user-specified terminal time under dual saturation. The outer loop employs a constrained optimal control problem to generate a reference trajectory that respects dual-saturation constraints, while the inner loop uses a robust prescribed-time prescribed-accuracy controller to track the reference within the desired accuracy by the terminal time. Through this decoupled architecture, the proposed framework effectively overcomes the fundamental shortcomings of

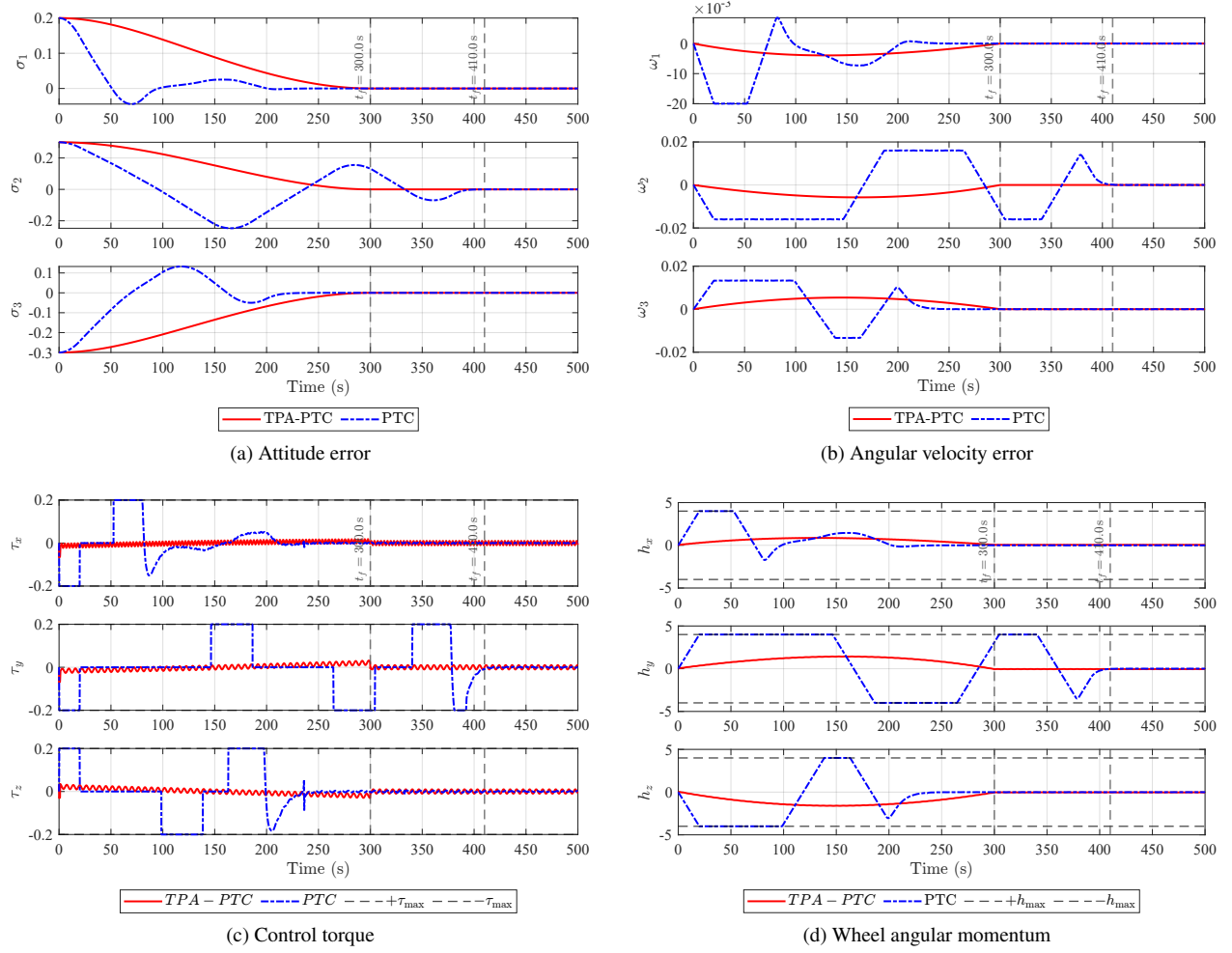


Fig. 8 Case 3.2: Performance under dual saturation

existing PTC methods, namely the inherent conservatism and insufficient consideration of dual saturation. Furthermore, by identifying the minimum feasible maneuver time $T_{f,\min}$ and establishing a parameter-synthesis methodology, the proposed framework significantly strengthens engineering realizability while streamlining the design process. Numerical simulations validated the effectiveness of the proposed framework in achieving exact terminal-time convergence, tunable precision, and dual-saturation compliance. Future work will focus on multi-spacecraft cooperative maneuvers.

A. Estimate of Minimum Feasible Maneuver Time

This appendix provides a practical estimate of $T_{f,\min}$ for use in Assumption 1. Given initial attitude $\sigma_e(0)$, the principal rotation angle θ and axis $\mathbf{e}$ satisfy $\theta = 4 \arctan(\|\sigma_e(0)\|)$ and $\mathbf{e} = \sigma_e(0)/\|\sigma_e(0)\|$ (or any unit vector if $\sigma_e(0) = \mathbf{0}$). Define $J_e = \mathbf{e}^\top J_0 \mathbf{e}$. Under component-wise torque limits $|\tau_i| \leq \tau_{i,\max}$, the maximum torque projection along $\mathbf{e}$ is $\tau_{e,\max} = \sum_{i=1}^3 |e_i| \tau_{i,\max}$.

Assuming $\mathbf{h}_w(0) = \mathbf{0}$ and a bang-bang allocation $\tau_i = \text{sgn}(e_i) \tau_{i,\max}$, the maximum duration before momentum

saturation is $t_h = \min_i h_{w,i,\max}/\tau_{i,\max}$. For a rest-to-rest rotation about $\mathbf{e}$, the peak rate is $\omega_{\max} = \tau_{e,\max} t_h / J_e$ and the no-coast angle is $\theta_h = \tau_{e,\max} t_h^2 / J_e$. The minimum-time estimate is then

$$T_{f,\min} \approx \begin{cases} 2\sqrt{J_e \theta / \tau_{e,\max}}, & \theta \leq \theta_h, \\ 2t_h + (\theta - \theta_h) / \omega_{\max}, & \theta > \theta_h \end{cases} \quad (43)$$

B. Bound on the Feedforward Mismatch

The reference angular momentum $\mathbf{H}_d = \mathbf{J}_0 \boldsymbol{\omega}_d + \mathbf{h}_{w,d}$, generated under disturbance-free dynamics from $\mathbf{H}_d(0) = \mathbf{H}(0) = \mathbf{H}_0$, satisfies $\|\mathbf{H}_d(t)\| = H_0$ and $\|\mathbf{H}_d(t) - \mathbf{H}_0\| \leq 2H_0$. The reference torque is $\boldsymbol{\tau}_{\text{ref}} = \mathbf{J}_0 \dot{\boldsymbol{\omega}}_d + \boldsymbol{\omega}_d^\times \mathbf{H}_d$. Substituting $\mathbf{f}$ from (6) into $\Delta_f = -\mathbf{f} - \boldsymbol{\tau}_{\text{ref}}$ and simplifying,

$$\begin{aligned} \Delta_f &= \boldsymbol{\omega}^\times (\mathbf{H} - \mathbf{H}_0) + \boldsymbol{\omega}_e^\times \mathbf{H}_0 + [(\mathbf{R} - \mathbf{I})\boldsymbol{\omega}_d]^\times \mathbf{H}_0 + \boldsymbol{\omega}_d^\times (\mathbf{H}_0 - \mathbf{H}_d) \\ &\quad - \boldsymbol{\omega}^\times \Delta \mathbf{J} \boldsymbol{\omega} - \mathbf{J}_0 \boldsymbol{\omega}_e^\times \mathbf{R} \boldsymbol{\omega}_d + \mathbf{J}_0 (\mathbf{R} - \mathbf{I}) \dot{\boldsymbol{\omega}}_d \end{aligned} \quad (44)$$

Invoking $\|\mathbf{H} - \mathbf{H}_0\| \leq H_{\max}$ (Assumption 3), $\|\mathbf{H}_0 - \mathbf{H}_d\| \leq 2H_0$, $\|\mathbf{I} - \mathbf{R}\| \leq 4\|\boldsymbol{\sigma}_e\|$, and $\|\mathbf{a}^\times \mathbf{b}\| \leq \|\mathbf{a}\| \|\mathbf{b}\|$, the component-wise bound is

$$\begin{aligned} |\Delta_{f,i}| &\leq (\|\boldsymbol{\omega}_e\| + \|\boldsymbol{\omega}_d\|) H_{\max} + \|\boldsymbol{\omega}_e\| H_0 + 4\|\boldsymbol{\sigma}_e\| \|\boldsymbol{\omega}_d\| H_0 + 2\|\boldsymbol{\omega}_d\| H_0 \\ &\quad + \hat{\nu}_i (\|\boldsymbol{\omega}_e\| + \|\boldsymbol{\omega}_d\|)^2 + \mu_i \|\boldsymbol{\omega}_e\| \|\boldsymbol{\omega}_d\| + 4\mu_i \|\boldsymbol{\sigma}_e\| \|\dot{\boldsymbol{\omega}}_d\| \end{aligned} \quad (45)$$

where $\mu_i = \sum_{j=1}^3 |(J_0)_{ij}|$, $\nu_i = \sum_{j=1}^3 |(\Delta J)_{ij}|$, $\hat{\nu}_i = \sum_{j \neq i} \nu_j$. Substituting the supremum bounds $\|\boldsymbol{\omega}_e\| \leq \omega_{e,\max}$, $\|\boldsymbol{\sigma}_e\| \leq \sigma_{\max}$, $\|\boldsymbol{\omega}_d\| \leq \omega_{d,\max}$, $\|\dot{\boldsymbol{\omega}}_d\| \leq \dot{\omega}_{d,\max}$ yields the envelope $\Delta_{f,i,\max}$ used in Theorem 4.

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References

- [1] Alex Pothén, A., Crain, A., and Ulrich, S., “Pose Tracking Control for Spacecraft Proximity Operations Using the Udwadia–Kalaba Framework,” *Journal of Guidance, Control, and Dynamics*, Vol. 45, No. 2, 2022, pp. 296–309. <https://doi.org/10.2514/1.G005169>, URL <https://doi.org/10.2514/1.G005169>, publisher: American Institute of Aeronautics and Astronautics _eprint: <https://doi.org/10.2514/1.G005169>.
- [2] Das, G., and Sinha, M., “Predefined-Time Hybrid Tracking Control of Flexible Satellites with Conditional Anti-Unwinding,” *Journal of Guidance, Control, and Dynamics*, Vol. 47, No. 11, 2024, pp. 2426–2434. <https://doi.org/10.2514/1.G008286>, URL <https://doi.org/10.2514/1.G008286>.
- [3] Xu, C., Zelazo, D., and Wu, B., “Distributed prescribed-time coordinated control of spacecraft formation flying under input saturation,” *Advances in Space Research*, Vol. 74, No. 5, 2024, pp. 2302–2315. <https://doi.org/10.1016/j.asr.2024.05.077>, URL <https://www.sciencedirect.com/science/article/pii/S027311772400543X>.
- [4] Xiao, Y., J. de Ruiter, A. H., Ye, D., and Sun, Z., “Attitude Tracking Control for Rigid-Flexible Coupled Spacecraft with Guaranteed Performance Bounds,” *Journal of Guidance, Control, and Dynamics*, Vol. 43, No. 2, 2020, pp. 327–337. <https://doi.org/10.2514/1.G004535>.
- [5] Ye, D., Zou, A.-M., and Sun, Z., “Predefined-Time Predefined-Bounded Attitude Tracking Control for Rigid Spacecraft,” *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 58, No. 1, 2022, pp. 464–472. <https://doi.org/10.1109/TAES.2021.3103258>, URL <https://ieeexplore.ieee.org/abstract/document/9512481>.
- [6] Xiao, Y., Yang, Y., Ye, D., and Zhang, J., “Quantitative Precision Second-Order Temporal Transformation-Based Pose Control for Spacecraft Proximity Operations,” *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 61, No. 2, 2025, pp. 1931–1941. <https://doi.org/10.1109/TAES.2024.3469167>.
- [7] Hong, Y., and Jiang, Z.-P., “Finite-Time Stabilization of Nonlinear Systems With Parametric and Dynamic Uncertainties,” *IEEE Transactions on Automatic Control*, Vol. 51, No. 12, 2006, pp. 1950–1956. <https://doi.org/10.1109/TAC.2006.886515>, URL <https://ieeexplore.ieee.org/document/4026648/>.

- [8] Zhao, L., Yu, J., and Yu, H., “Adaptive Finite-Time Attitude Tracking Control for Spacecraft With Disturbances,” *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 54, No. 3, 2018, pp. 1412–1421. <https://doi.org/10.1109/TAES.2017.2764472>.
- [9] Shi, X.-N., Zhou, Z.-G., and Zhou, D., “Finite-Time Attitude Trajectory Tracking Control of Rigid Spacecraft,” *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 53, No. 6, 2017, pp. 2913–2923. <https://doi.org/10.1109/TAES.2017.2710398>.
- [10] Polyakov, A., “Nonlinear Feedback Design for Fixed-Time Stabilization of Linear Control Systems,” *IEEE Transactions on Automatic Control*, Vol. 57, No. 8, 2012, pp. 2106–2110. <https://doi.org/10.1109/TAC.2011.2179869>, URL <https://ieeexplore.ieee.org/document/6104367/>.
- [11] Polyakov, A., Efimov, D., and Perruquetti, W., “Finite-time and fixed-time stabilization: Implicit Lyapunov function approach,” *Automatica*, Vol. 51, 2015, pp. 332–340. <https://doi.org/10.1016/j.automatica.2014.10.082>, URL <https://www.sciencedirect.com/science/article/pii/S0005109814004634>.
- [12] del Valle, S. C., Urrutxua, H., and Solano-López, P., “Generalized Potter–Stern Algorithm for Impulsive Minimum-Fuel Fixed-Time Rendezvous,” *Journal of Guidance, Control, and Dynamics*, Vol. 48, No. 12, 2025, pp. 2744–2762. <https://doi.org/10.2514/1.G009164>, URL <https://doi.org/10.2514/1.G009164>.
- [13] Sánchez-Torres, J. D., Sanchez, E. N., and Loukianov, A. G., “Predefined-time stability of dynamical systems with sliding modes,” *2015 American Control Conference (ACC)*, 2015, pp. 5842–5846. <https://doi.org/10.1109/ACC.2015.7172255>, URL <https://ieeexplore.ieee.org/document/7172255>, iSSN: 2378-5861.
- [14] Song, Y., Wang, Y., Holloway, J., and Krstic, M., “Time-varying feedback for regulation of normal-form nonlinear systems in prescribed finite time,” *AUTOMATICA*, Vol. 83, 2017, pp. 243–251. <https://doi.org/10.1016/j.automatica.2017.06.008>, URL <https://linkinghub.elsevier.com/retrieve/pii/S000510981730290X>, num Pages: 9 Place: Oxford Publisher: Pergamon-Elsevier Science Ltd Web of Science ID: WOS:000408288800028.
- [15] Sánchez-Torres, J. D., Gómez-Gutiérrez, D., López, E., and Loukianov, A. G., “A class of predefined-time stable dynamical systems,” *IMA Journal of Mathematical Control and Information*, Vol. 35, No. Supplement_1, 2018, pp. i1–i29. <https://doi.org/10.1093/imamci/dnx004>, URL <https://doi.org/10.1093/imamci/dnx004>.
- [16] Xiao, Y., Wang, Y., Ye, D., and Sun, Z., “Predefined-time robust attitude tracking control of flexible spacecraft with continuous and nonsingular performance,” *Aerospace Science and Technology*, Vol. 159, 2025, p. 110000. <https://doi.org/10.1016/j.ast.2025.110000>, URL <https://www.sciencedirect.com/science/article/pii/S1270963825000720>.
- [17] Zhou, B., “Finite-time stability analysis and stabilization by bounded linear time-varying feedback,” *Automatica*, Vol. 121, 2020, p. 109191. <https://doi.org/10.1016/j.automatica.2020.109191>, URL <https://www.sciencedirect.com/science/article/pii/S0005109820303897>.

- [18] Cao, Y., Cao, J., and Song, Y., “Practical prescribed time tracking control over infinite time interval involving mismatched uncertainties and non-vanishing disturbances,” *Automatica*, Vol. 136, 2022, p. 110050. <https://doi.org/10.1016/j.automatica.2021.110050>, URL <https://www.sciencedirect.com/science/article/pii/S0005109821005781>.
- [19] Xiao, Y., Yang, Y., Ye, D., and Zhao, Y., “Scaling-Transformation-Based Attitude Tracking Control for Rigid Spacecraft With Prescribed Time and Prescribed Bound,” *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 61, No. 1, 2025, pp. 433–442. <https://doi.org/10.1109/TAES.2024.3451454>.
- [20] Sun, H.-J., Wu, Y.-Y., and Zhang, J., “Attitude synchronization control for multiple spacecraft: A preassigned finite-time scheme,” *Advances in Space Research*, Vol. 73, No. 12, 2024, pp. 6094–6110. <https://doi.org/10.1016/j.asr.2024.02.001>, URL <https://www.sciencedirect.com/science/article/pii/S0273117724001303>.
- [21] Xu, C., Wu, B., and Wang, D., “Distributed Prescribed-Time Attitude Coordination for Multiple Spacecraft With Actuator Saturation Under Directed Graph,” *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 58, No. 4, 2022, pp. 2660–2672. <https://doi.org/10.1109/TAES.2021.3135484>, URL <https://ieeexplore.ieee.org/document/9650527/>.
- [22] Zhao, S., Zheng, J., Yi, F., Wang, X., and Zuo, Z., “Exponential Predefined Time Trajectory Tracking Control for Fixed-Wing UAV With Input Saturation,” *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 60, No. 5, 2024, pp. 6406–6419. <https://doi.org/10.1109/TAES.2024.3402656>, URL <https://ieeexplore.ieee.org/document/10534850/>.
- [23] Wallsgrove, R. J., and Akella, M. R., “Globally Stabilizing Saturated Attitude Control in the Presence of Bounded Unknown Disturbances,” *Journal of Guidance, Control, and Dynamics*, Vol. 28, No. 5, 2005, pp. 957–963. <https://doi.org/10.2514/1.9980>.
- [24] Hu, Q., Tan, X., and Akella, M. R., “Finite-Time Fault-Tolerant Spacecraft Attitude Control with Torque Saturation,” *Journal of Guidance, Control, and Dynamics*, Vol. 40, No. 10, 2017, pp. 2524–2537. <https://doi.org/10.2514/1.G002191>.
- [25] Miller, A. J., and Akella, M. R., “Guidance Templates for Spacecraft Attitude Maneuver Planning,” *Journal of Guidance, Control, and Dynamics*, Vol. 47, No. 7, 2024, pp. 1470–1478. <https://doi.org/10.2514/1.G007862>.
- [26] Kobayashi, H., Shoji, Y., and Yamada, K., “Optimal Attitude Trajectory of Spacecraft with Pyramid-Type CMGs Using a Pseudo-Spectral Method,” *AEROSPACE TECHNOLOGY JAPAN, THE JAPAN SOCIETY FOR AERONAUTICAL AND SPACE SCIENCES*, Vol. 16, No. 0, 2017, pp. 55–63. <https://doi.org/10.2322/astj.16.55>, URL https://www.jstage.jst.go.jp/article/astj/16/0/16_JSASS-D-16-00034/_article/-char/ja/.
- [27] Spiller, D., Melton, R. G., and Curti, F., “Inverse dynamics particle swarm optimization applied to constrained minimum-time maneuvers using reaction wheels,” *Aerospace Science and Technology*, Vol. 75, 2018, pp. 1–12. <https://doi.org/10.1016/j.ast.2017.12.038>, URL <https://www.sciencedirect.com/science/article/pii/S1270963817308684>.
- [28] Celani, F., and Lucarelli, D., “Spacecraft Attitude Motion Planning Using Gradient-Based Optimization,” *Journal of Guidance, Control, and Dynamics*, Vol. 43, No. 1, 2020, pp. 140–145. <https://doi.org/10.2514/1.G004531>, URL <https://arc.aiaa.org/doi/10.2514/1.G004531>, publisher: American Institute of Aeronautics and Astronautics.

- [29] Fan, Y., Xu, R., Li, Z., Zhu, S., Shang, H., Gao, A., and Gao, Y., “Polynomial Attitude Trajectory Planning for Spacecraft with Movable Parts Using Decoupled Strategy,” *Journal of Guidance, Control, and Dynamics*, Vol. 47, No. 1, 2024, pp. 72–86. <https://doi.org/10.2514/1.G007645>, URL <https://doi.org/10.2514/1.G007645>.
- [30] Mammarella, M., Lee, D. Y., Park, H., Capello, E., and Ghiglino, P., “Attitude Control of a Small Spacecraft via Tube-Based Model Predictive Control,” *Journal of Spacecraft and Rockets*, Vol. 56, No. 6, 2019, pp. 1585–1597. <https://doi.org/10.2514/1.A34394>.
- [31] Yang, Y., “Attitude Model Predictive Control with Actuator Saturation Using an Arc-Search Interior-Point Method,” *Journal of Guidance, Control, and Dynamics*, Vol. 46, No. 5, 2023, pp. 969–978. <https://doi.org/10.2514/1.G007179>.
- [32] Xiao, Y., Yang, Y., Ye, D., and Zhao, Y., “Scaling-Transformation-Based Attitude Tracking Control for Rigid Spacecraft With Prescribed Time and Prescribed Bound,” *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 61, No. 1, 2025, pp. 433–442. <https://doi.org/10.1109/TAES.2024.3451454>, URL <https://ieeexplore.ieee.org/document/10659135>.
- [33] Anguiano-Gijón, C. A., Muñoz-Vázquez, A. J., Sánchez-Torres, J. D., Romero-Galván, G., and Martínez-Reyes, F., “On predefined-time synchronisation of chaotic systems,” *Chaos, Solitons & Fractals*, Vol. 122, 2019, pp. 172–178. <https://doi.org/10.1016/j.chaos.2019.03.015>, URL <https://www.sciencedirect.com/science/article/pii/S0960077919300864>.
- [34] Patterson, M. A., and Rao, A. V., “GPOPS-II: A MATLAB Software for Solving Multiple-Phase Optimal Control Problems Using hp-Adaptive Gaussian Quadrature Collocation Methods and Sparse Nonlinear Programming,” *ACM Trans. Math. Softw.*, Vol. 41, No. 1, 2014, pp. 1:1–1:37. <https://doi.org/10.1145/2558904>, URL <https://dl.acm.org/doi/10.1145/2558904>.